

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

Chasing Control Faults in a Corrosive Plant

For maintenance techs and plant engineers at composting, organics and anaerobic digestion (AD) sites who keep getting intermittent control faults that nobody can pin down. This guide helps you tell an environment-driven instrument fault apart from a real process problem, before you spend another shift on it.

How to use this

Work it symptom-first. Find the row that matches what you're actually seeing, run the checks in order, and write down what you find. The single most useful thing you can do on an intermittent fault is log it: time, weather, what the process was doing, and the exact reading. Half the corrosive-plant faults I get called on resolve themselves the moment somebody has three logged occurrences to look at instead of one story. Keep this near the panel, not in a binder in the office.

A note on why this plant is different. In a clean packaging line, a signal problem is usually a signal problem. In a plant that makes hydrogen sulfide (H₂S), ammonia (NH₃) and warm wet air all day, the environment is a slow, active participant. H₂S plus moisture attacks copper and silver: it turns bright terminations dull, then black, then powdery. Ammonia goes after copper too, and loves to creep into any enclosure that breathes. Warm humid process air hits a cooler panel overnight and condenses inside it. Dust from screening and grinding coats boards and bridges terminals once it gets damp. None of that shows up as a clean, repeatable fault. It shows up as drift, as intermittents that track the weather, and as things that work fine until you touch them.

Before you open a box or break a live loop

Read this first — several rows below have you opening junction boxes and metering an energized loop in an atmosphere that can be both toxic and explosive.

- **Confirm the hazardous-area (explosive-atmosphere) classification for the exact spot you're standing in, and whether a hot-work or energized-work permit is required, before you open an enclosure or break a live loop.** In a plant making methane and H₂S, opening a box or making and breaking a live connection can be an ignition source. That's a permit question, not a call you make alone at the panel.

- **Wear a personal H₂S and lower-explosive-limit (LEL) gas monitor and don't trust your nose.** H₂S deadens your sense of smell as it climbs; by the time a concentration is dangerous you may not smell it at all.
- **Intrinsically-safe circuits must not be metered or broken with non-approved gear.** Putting an ordinary meter or clamp on an IS loop, or breaking it with uncertified tools, defeats the very protection that makes energized work in that area safe.

Symptom → likely causes → what to check / what to log

Analog signals (4-20 mA loops, resistance temperature detectors (RTDs), thermocouples)

Symptom	Likely causes	What to check / what to log
Reading slowly drifts up over weeks, then jumps back after someone reseats a connector	High-resistance joint from sulfide film on terminals or a corroding splice; moisture in a junction box creating a leakage path to ground	For a suspected series joint, put a meter across the suspect terminals or splice while the loop is energized and read the millivolt drop under load: more than a few millivolts across what should be a clean termination is a filmed or corroded joint. (In a series current loop the current is identical everywhere, so a resistive joint shows up as voltage drop and lost compliance, not a current difference.) To catch the moisture case, compare loop current at the transmitter against current at the input card: a shortfall at the card means current is leaking off to ground somewhere in between. Reading loop current at two points means breaking the loop in series or using a true DC milliamp clamp or loop calibrator — an ordinary AC clamp meter won't read a DC loop, and the number on a HART communicator or the transmitter's own display is not the measured analog loop current. Open field junction boxes and look for green/black powder, white ammonia salts, or water tracks. Log the drift rate and whether reseating fixes it and for how long.
Erratic or noisy signal that gets worse on humid nights or after washdown	Moisture ingress at a gland or J-box; degraded shield ground; condensation inside a transmitter housing	Check for water in the bottom of enclosures and glands pointing upward. Confirm the cable shield is grounded at one end only. Log time of day, humidity, and whether it correlates with washdown or ambient swings.

Symptom	Likely causes	What to check / what to log
RTD or thermocouple reads high, low, or open intermittently	Corroded RTD lead terminations; thermocouple extension wire attacked at a terminal block; wet connector	Wiggle-test at the head and at the card with the loop live and watch the reading. Inspect the sensor head — these fill with condensate. Log which channel, and whether a spare channel on the same card behaves.
Whole card or several channels read off together	Common ground or reference corroding; power supply issue; card itself attacked by gas	Compare several channels' behavior. Check the card's common/ground terminals for film. Log whether it's one channel or a group; a group points upstream of the field wiring.

Digital I/O, relays and connectors

Symptom	Likely causes	What to check / what to log
Input that should be steady chatters or flickers	Sulfide film on a dry contact or terminal raising contact resistance; damp dust bridging terminals	Meter across the field contact when it's supposed to be closed: a fraction of a volt drop across a "closed" contact is a filmed contact. Log which input and what the field device is.
Relay or contactor that used to pull in clean now buzzes or is unreliable	Corroded coil terminations; pitted or filmed contacts; moisture in the enclosure	Inspect contacts for black film and pitting. Check coil voltage at the coil, not at the source. Log operations count if you have it and whether it's one device or several of the same type.
Connector-based fault that comes and goes when the panel is opened or bumped	Edge connectors, ribbon headers and pluggable terminal blocks growing sulfide film on the mating surfaces	This is the classic. If touching it changes it, the connector is the suspect. Unplug and reseat a suspect connector; if it clears for days then returns, that's film reforming. Log the connector and the return interval.
Communication drops on a fieldbus or network segment (any protocol)	Corroded connector pins or shield drain; moisture in an RJ45 or M12; a damaged run acting up when wet	Check error/retry counters at both ends if the controller exposes them. Inspect connectors for green pins. Log which segment, which drops, and whether it tracks humidity or rain.

Sensors and analyzers exposed to the process

Symptom	Likely causes	What to check / what to log
Gas reading (H ₂ S, methane, oxygen) drifts or reads low	Sensor at or past end of life in a high-concentration environment; blocked or fouled sample path; filter caked with dust or moisture	Check last calibration and bump-test date. Inspect sample lines and filters. Do not troubleshoot a gas-detection channel by disabling it — see the safety note. Log reading, calibration date, and sensor age.
pH, dissolved oxygen or similar process analyzer wandering	Coated or fouled probe; reference junction poisoned by sulfide; wet or corroded connector at the head	Clean and recalibrate per the maker's procedure before you blame electronics. Inspect the connector at the sensor. Log whether cleaning restores it and how fast it drifts back.
Level, pressure or flow transmitter reading unstable	Condensation or corrosion in the housing; plugged impulse line or reference leg (common in wet, particulate service); diaphragm coating	Separate the sensing element from the electronics: does the raw process condition match an independent check? Log the reading against a manual or redundant measurement.

A decision sequence for the intermittent one

Intermittents are where people lose days. Work it like this.

Start here: is the fault at the field device, or in the wiring and panel between it and the controller? Read the value as close to the source as you can and again at the input card. If they disagree, the problem is in between — wiring, glands, junction boxes, terminals. If they agree and both are wrong, the field device or the process is the issue.

If it moves when you touch it, it's mechanical or a connection: filmed terminals, a loose or corroded connector, a wet gland. Reseat one suspect at a time and note how long the fix holds. A fix that lasts days and then fails again is film reforming — clean properly, don't just reseat.

If it tracks the weather or time of day, think moisture and condensation. Faults that appear on humid nights, clear by mid-morning, or follow washdown are almost always ingress or condensation, not the electronics failing.

If it's steady but wrong, and reads wrong everywhere including at the source, now you have a genuine candidate for a real process problem or a truly dead sensor, not an environment artifact. This is the important fork: don't chase a process ghost that's actually a corroded terminal, and don't keep cleaning connectors when the tank really is at that level.

Cross-check before you commit. One independent measurement tells you whether the instrument is lying or the process really moved: a handheld meter, a redundant transmitter, a manual reading, a sight glass. That single check has saved more callouts than any tool in the bag.

| Enclosure, purge and gland practices worth checking

- **Glands point down, and are actually tightened on the right cable diameter.** An upward-facing or oversized gland is an open invitation for water, and it's the most common ingress point I find.
- **Unused gland holes and knockouts are sealed.** Corrosive air finds every gap.
- **The enclosure breathes as little as possible.** Every cycle of warm wet air in and cool air out deposits moisture inside. A well-sealed enclosure with its rating intact beats a "sealed" one with three open knockouts.
- **If the panel is purged or pressurized, confirm it's actually holding positive pressure** and the air source is clean and dry. A failed purge is worse than none because people trust it. Never treat a purge or pressurization system tied to a safety function as something to defeat when it nuisance-alarms; find out why it dropped.
- **Check corrosion inhibitors and whether anyone's replacing them.** Vapor-phase inhibitor sachets and coated-board treatments help, but they have a service life and get ignored for years.
- **Heat kills condensation.** A correctly sized enclosure heater, or the panel's own load keeping the interior above dew point, prevents a lot of overnight condensation. Verify it's working, not just installed.

None of this is exotic. It's the stuff that quietly stops being maintained after commissioning, and the plant environment punishes that within a couple of seasons.

| 10-minute triage checklist

Run this before you call anyone. It catches a surprising share of these faults and, when it doesn't, it gives whoever you call something real to work with.

- ☐ Write down the exact reading or fault, the time, and the weather (temperature, humidity, rain, recent washdown).
- ☐ Open the nearest junction box and enclosure and look: standing water, green/black powder on copper, white salts, dust bridging terminals, glands facing up.
- ☐ Read the signal at the source and again at the input card. Do they agree?
- ☐ Gently wiggle-test terminations and connectors with the loop live; note anything that changes the reading.
- ☐ Meter the voltage drop across any "closed" field contact that's acting up.
- ☐ Take one independent cross-check of the value (handheld, redundant device, manual reading).

- ☐ For any comms fault, check error/retry counters at both ends and inspect the connectors.
- ☐ Confirm you have not touched, bypassed, or forced anything tied to gas detection or a safety interlock.

If two or three of these point the same way, you probably have your answer. If they scatter, you have data worth handing to someone.

| Safety note — gas detection and interlocks

Gas detection, ventilation interlocks and protective trips are what keep this plant survivable, so never bypass, force, jumper or disable a gas-detection channel or a safety interlock to clear a fault or keep production running. A gas sensor reading wrong gets fixed or calibrated on its proper schedule; you don't switch it off. Retuning a genuine nuisance trip so it stops false-tripping is legitimate work when the protection still functions afterward, but jumpering the protection out so nobody has to look at it puts people at risk. If you're not certain which side of that line you're on, stop and get someone who is, and follow your site's lockout/tagout and confined-space procedures every time.

| Honest limits

This covers the environment-driven faults I see most in wet, corrosive organics plants, not genuine control-logic bugs, corrosion-unrelated hardware failures, or process problems that only look electrical — and since your gas concentrations, washdown regime and installed brands all move the odds, treat it as a way to gather evidence and narrow the field, not a guarantee.

| If it keeps getting handed around

The corrosive-plant faults that eat the most time are the ones bounced between the sensor vendor, the panel builder and the integrator while the intermittent comes back every humid week. If you'd rather have a vendor-neutral engineer look at the whole loop at once, I'm happy to talk it through. No product to sell you either way.

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